# SITHUM MANUSHA

**SOFTWARE ENGINEERING INTERN**

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## ABOUT ME

Third-year undergraduate studying Information Technology and Management at the University of Moratuwa, passionate about modern web technologies and software development. Experienced in developing frontend interfaces and backend APIs using React, Next.js, Node.js, and Java Spring Boot. Seeking a Software Engineering Internship where I can apply my problem-solving skills, collaborate on real-world projects, and develop professionally.

## EDUCATION

**University of Moratuwa (UOM) - Moratuwa, Sri Lanka** **2024 - Present**
B.Sc. (Hons) in Information Technology & Management **(3rd Year | CGPA: 3.23 / 4.0)**

**Open University of Sri Lanka (OUSL) - Nawala, Sri Lanka** **2025 - Present**
Bachelor of Laws Honours Degree Programme (LL.B.)

**St. Joseph's Vaz College - Wennappuwa, Sri Lanka** **2009 - 2022**
G.C.E. Advanced Level (2 A's & 1 B) | G.C.E. Ordinary Level (8 A's & 1 B)

## FEATURED PROJECTS

**1. Super POS — Multi-Branch Enterprise Offline-First Retail System (Group Project)** **2026**

- Built a client-side offline sync queue (**useOfflineSync** with **IndexedDB offlineQueue**) and batch retry pipelines (**/sync/push, /sync/retry, /sync/batch-retry**) to buffer offline sales, stock updates, and price changes with auto-reconnection sync to **PostgreSQL**.
- Implemented **Last-Write-Wins (LWW)** conflict detection with an interactive 3-way resolver (**Trust Local, Trust Cloud, and Smart Merge**) and **Recharts** analytics dashboards tracking P50/P95/P99 latency and sync throughput.
- Developed a live regional node topology monitor and fleet broadcast system via **Socket.io WebSockets**, alongside remote device access revocation, entity sync policies, and an audit trail with **PDF health report exports**.
- **Technologies:** Next.js, React, TypeScript, Tailwind CSS, IndexedDB, Recharts, NestJS, Node.js, Prisma ORM, Socket.io, Zod, PostgreSQL (Supabase), Turborepo
- **Live Demo:** pos-system-pos-web-app.vercel.app | **GitHub:** github.com/Ryzera/Ryzera-POS

**2. CareerFlow — Full-Stack ATS Resume Optimizer & AI Interview Platform (Individual Project)** **2026**

- Built a full-stack platform with a **Java 21 Spring Boot 3** gateway, **Python 3.12 FastAPI** NLP worker, and **Next.js 14** client, containerized with **Docker Compose** and automated with **GitHub Actions CI/CD**.
- Developed an automated ATS resume scanner using **TF-IDF vectorization & Cosine Similarity** to detect missing keywords, audit format health, and rewrite bullets.
- Created an interactive 5-round AI mock interview simulator with API rate limiting (10 req/min) and **26 automated tests (JUnit 5, Pytest, Playwright)**.
- **Technologies:** Java 21, Spring Boot 3, Python 3.12, FastAPI, Next.js 14, PostgreSQL, Docker Compose, GitHub Actions, Scikit-learn, JUnit 5, Pytest, Playwright
- **Live Demo:** careerflow-app.onrender.com | **GitHub:** github.com/SithumManusha/careerflow

**3. Ledgerly — Personal Finance & Shared Expense Platform (Individual Project)** **2026**

- Built an AI-assisted financial tracker that monitors monthly expenses, calculates category budget health scores, and alerts users when spending limits are reached.
- Developed an interactive balance projector using **Recharts** to visualize 6-month savings trends based on customizable monthly contributions and expense adjustments.
- Implemented group expense splitting across 4 distribution methods with automated PDF statement exports, backed by a type-safe stack (**React, TypeScript, Node.js, PostgreSQL**) and **60+ automated unit tests (Vitest)**.
- **Technologies:** React, TypeScript, Node.js, Express, PostgreSQL, Drizzle ORM, Recharts, Vitest, Render
- **Live Demo:** ledgerly-mbcd.onrender.com | **GitHub:** github.com/SithumManusha/Ledgerly

**4. Automated Robotic Sorting System — Microcontroller Conveyor Platform (Group Project)** **2025**

Dual-Microcontroller Mechatronics & RFID Distribution

- Constructed an automated conveyor sorting prototype equipped with a 4-DOF robotic arm, ultrasonic sensors, an RFID reader, and 3 sorting compartments.
- Engineered a two-finger mechanical gripper driven by an Arduino and stepper motor, calibrating motor step counts to achieve 98% pick-and-place precision across 3 RFID object categories.
- Tested sensor signal timings and synchronized robotic arm pick-and-place routines with continuous conveyor belt movement.
- **Technologies:** Arduino Mega/Uno (C++), Stepper Motors (NEMA 17, 28BYJ-48), ULN2003, Servo Motors, Integration Testing
- **GitHub:** github.com/SithumManusha/robotic-sorting-system

## TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, C, SQL
- **Front-end Development:** React, Next.js 14/16, TypeScript, Tailwind CSS, Recharts, IndexedDB, State Management
- **Back-end Development:** Spring Boot 3, FastAPI, Node.js, Express.js, NestJS, Socket.io, Zod, RESTful APIs
- **Databases & ORM:** PostgreSQL (Supabase), MySQL, Redis, Spring Data JPA, Hibernate, Prisma ORM, Drizzle ORM
- **Testing & Quality:** JUnit 5, Pytest, Playwright (E2E), Vitest, Postman API Testing, Jira
- **DevOps & Tools:** Docker, Docker Compose, Turborepo, Git, GitHub Actions (CI/CD), Postman, Swagger / OpenAPI, Render Cloud

## CORE COMPETENCIES & ARCHITECTURE

Microservices Architecture, RESTful API Design, Agile / Scrum, Object-Oriented Design (SOLID), Clean Code, Automated Testing, Offline-First Sync

## CERTIFICATIONS

- AWS Cloud Practitioner Essentials – Amazon Web Services (AWS)
- Software Testing with Generative AI – Simplilearn
- Introduction to Artificial Intelligence – Simplilearn
- Introduction to Cybersecurity – Cisco Networking Academy

## LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Member of the Organizing Committee, Official Career Fair conducted by University of Moratuwa (2026)
- Member of the Organizing Committee, Project "Rest with Hope" conducted by Leo Club, UoM (2025)
- Active Member – FIT Moments (Faculty Media Body) | Leo Club of UoM | Legal Youth Community Sri Lanka

## LANGUAGES

- English: Professional Working Proficiency (Oral & Written) | Sinhala: Native Proficiency

## REFERENCES

**Dr. A.L.A.R.R. Thanuja (B.Sc. (Sri Lanka), Ph.D. (USA)),**
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